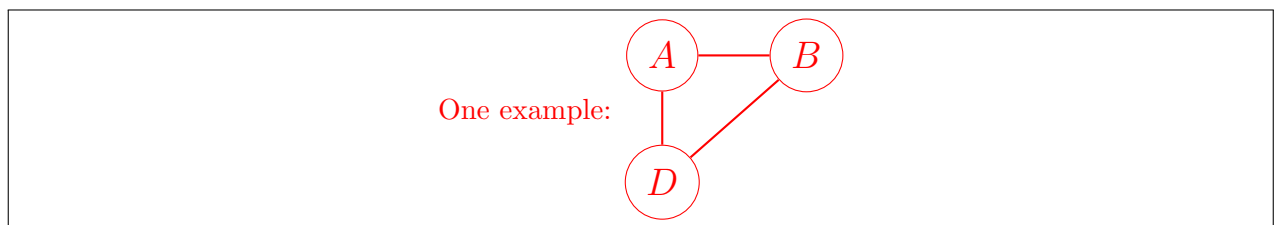


1. 25 points Draw a network which has exactly three spanning trees.



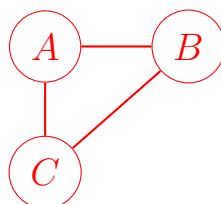
2. 25 points List some property/properties of a graph which are sufficient to categorize it as a network. List some property/properties of a graph which are sufficient to categorize it as a tree. Draw an example of each.

A network is a graph, G , which is connected (i.e. G is a graph in which there is a path between any two vertices in G).

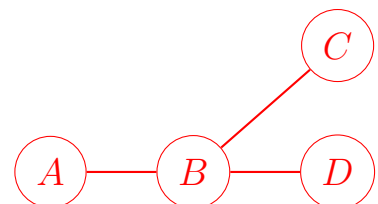
A tree, T , is a simple, connected graph (i.e. a simple network) which satisfies all of the following properties (each of which is equivalent – i.e. if you had any of the following (along with connected and simple, as we mentioned above), your answer is correct):

1. T has no circuits;
2. there is a unique path between any two vertices in T (i.e. there is one, and only one, path between any two vertices in T);
3. every edge in T is a bridge;
4. T has one less edge than it has vertices (i.e. if T has n vertices, it must have $n - 1$ edges);
5. T is a minimally connected network (i.e. T has as few edges as possible while still being a connected graph).

One example of a network:



One example of a tree:



3. 25 points Explain how you would go about finding a MST in a given weighted network. If there is an algorithm which accomplishes this task, it is sufficient for you to state the name of that algorithm and describe the steps involved in using it.

To find a MST in a given weighted network, let's call it G , one can use Kruskal's algorithm. If G has N vertices, Kruskal's algorithm can be stated as follows:

- Step 1: Pick the cheapest edge available (in the case of a tie, pick one amongst the tied edges at random). Mark it.
- Step 2: Pick the second cheapest edge available (in the case of a tie, pick one amongst the tied edges at random). Mark it.
- Step 3 through $N-1$: Continue on, picking and marking the cheapest unmarked edge available that does not create a circuit among the other marked edges.

4. 25 points Give a real-world example of a weighted network and a real-world interpretation of a MST in that network.

See the lecture slides for chapter 7 for many real-world examples of weighted networks and MST's (recall, these slides can be found on the course's WyoCourses page under the 'files' tab in the 'lecture notes' folder).